COVID Mortality Disruption Lab

Which causes of death were most disrupted by the COVID-19 pandemic, how persistent were those disruptions, and how did they vary across U.S. counties?

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Executive summary

Of 6 major causes of death tested against real CDC WONDER mortality data (1999-2024), **5 show a statistically significant, still-unresolved deviation** from their pre-pandemic trend, four years after the pandemic began. Two results directly contradicted this project's own pre-registered priors: cancer, expected to show no disruption within this data window, showed a real (if modest) one; Alzheimer's disease, expected to show a large disruption, showed none. A negative-control validation, a three-axis sensitivity analysis, and full FDR correction across the test family were run before any result was reported, and every methodological deviation encountered along the way is logged rather than silently absorbed.

This design treats COVID-19 as a system-wide shock to the healthcare/public-health system, not as the disease under study, and asks what changed, how much, and where — not why, since mortality data alone cannot cleanly separate direct viral effects from deferred care, isolation, or economic-stress mechanisms. All results below are reported as associational, never causal, per the project's causal-language policy.

1. Research question

Which causes of death experienced the greatest and most statistically significant disruption during the COVID-19 pandemic (2020-2024), how persistent were those disruptions through the most recent available data, and how did disruption severity vary across U.S. counties by socioeconomic status, healthcare access, and rurality?

Six substantive causes were tested: diseases of heart, diabetes mellitus, Alzheimer's disease, cerebrovascular disease, drug overdose, and malignant neoplasms (cancer). A negative control (congenital malformations, a cause with no direct COVID mechanism) and a COVID-19 reference series (not a hypothesis test) round out the 8-series design. Every cause's confidence level was stated before any 2020-2024 data was pulled or analyzed, so results can be checked against stated priors rather than narrated after the fact.

2. Methods

Primary method: known-date interrupted time series ("excess mortality"). An expected trend is fit on 1999-2019 age-adjusted mortality rates, projected forward through 2020-2024 with a 95% prediction interval, and a year is flagged as significantly disrupted if the observed value falls outside that interval. The breakpoint (2020) is fixed by the shock's known date, not searched for.

Three-way persistence classification. For causes with a significant 2020-2021 disruption: Persisted (still significant, same direction, through 2024), Resolved (shrank back within the interval), or Reversed (flipped sign).

Independent cross-check. Three change-point methods (PELT, binary segmentation, segmented regression) run on each series to verify a breakpoint near 2020 without being told to. Disagreement here is not itself evidence against a result: the primary method tests one specific pre-registered date, while the cross-check methods search the whole series for whichever single breakpoint fits best.

Negative control. The identical pipeline run on a cause with no direct COVID mechanism. A significant result there would indicate the method is detecting an artifact, not a real signal — a hard gate, not a footnote.

Multiple-testing correction. Benjamini-Hochberg FDR correction applied across the 6 test causes as one family, separately from the heterogeneity-stage correction applied per cause across its context variables.

Data. CDC WONDER Underlying Cause of Death, two database vintages bridged at the 2018-2019 overlap: "1999-2020" (database D76) for the 1999-2019 baseline, "2018-2024, Single Race" (database D158) for the 2020-2024 post-shock period. County-level heterogeneity data (diabetes, drug overdose) covers pre-period 2015-2019 vs. post-period 2020-2024, regressed against real County Health Rankings & Roadmaps context variables.

3. Results

5 of 6 test causes show a significant disruption; **5 of 6** survive FDR correction.

Cause	Result	p-value	FDR-sig.	2020-21 dev.	2024 dev.
Drug overdose	Persisted	5.98e-08	Yes	+40.9%	-4.3%

Diabetes mellitus	Persisted	7.99e-07	Yes	+26.9%	+15.0%
Malignant neoplasms	Persisted	0.000194	Yes	+1.7%	+4.6%
Diseases of heart	Persisted	0.000633	Yes	+26.1%	+34.9%
Cerebrovascular disease	Persisted	0.00142	Yes	+37.0%	+57.3%
Alzheimer's disease	No significant disruption	0.807	No	+1.0%	-19.1%

"Deviation" is the effect size: how far the observed rate is from the expected pre-pandemic trend, as a percent. Significance and magnitude are different claims.

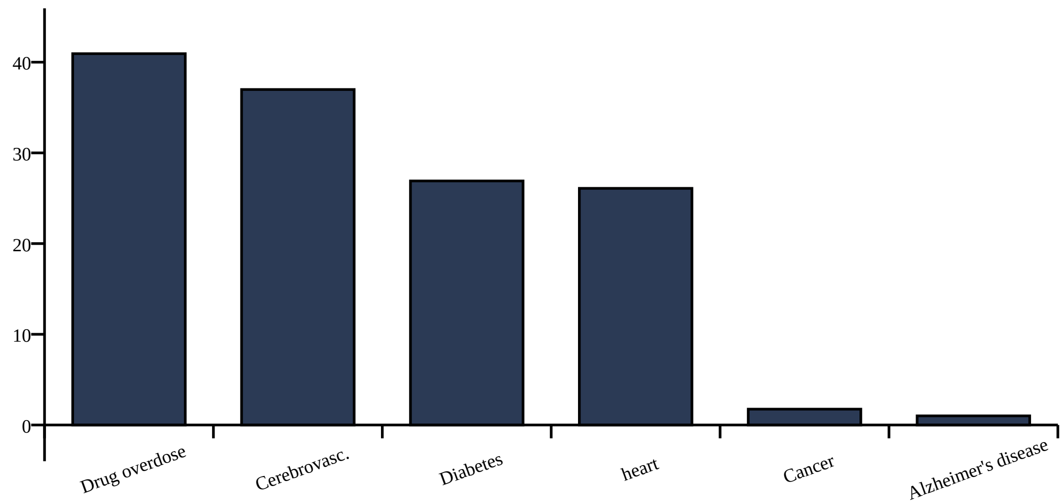


Figure 1. Mean 2020-2021 deviation from expected trend, by cause.

The two results that weren't supposed to happen this way

Cancer was expected to show nothing, and it didn't. The pre-registered prior was an explicit null result, with low confidence by design — delayed cancer screening and treatment during the pandemic was expected to take years longer than the 2024 data window to appear as excess mortality. Instead, cancer shows a persisted disruption (p = 0.000194, survives FDR correction), though its magnitude (+1.7% in 2020-21) is modest next to the larger disruptions above. The wider literature on this question is unsettled: early 2020 models projected large future increases in cancer deaths from delayed screening, but more recent modeling using actual pandemic-era England data found lung and breast cancer deaths came in lower than pre-pandemic trends predicted, not higher. This project's small, real, persisting effect should be read against that uncertainty.

Alzheimer's was expected to show a large effect, and it didn't. The pre-registered prior was high confidence of a large disruption, on the theory that pandemic-era isolation and care-facility disruption would show up clearly in dementia mortality. It didn't (p = 0.81). This doesn't mean isolation had no effect; it means that effect, if real, isn't visible in national mortality rates over this window using this method.

Negative control

Congenital malformations, deformations and chromosomal abnormalities — a cause concentrated in infancy with no direct COVID mechanism — shows no significant disruption, as expected (p = 0.682 on raw death counts, the gating metric; WONDER's 1-decimal age-adjusted-rate rounding made the rate-based test unreliable at this cause's low magnitude). This does not prove every positive result above is real, but a failure here would have been strong evidence the method was detecting an artifact.

Accidental drowning was the original negative control and failed — a real, statistically robust increase in deaths from 2020 onward, confirmed on raw counts, consistent with published CDC reporting on pandemic-era drowning increases (pool/beach closures, lifeguard shortages). It was swapped out because it was never actually COVID-independent, not because the method failed. Even the replacement isn't hermetically sealed from the pandemic: prenatal and obstetric care was also disrupted during 2020-2021, a real if smaller and more indirect pathway than the ones behind the 6 test causes.

4. Robustness

Three independent sensitivity checks re-fit the primary method one modeling choice at a time.

Baseline window (1999-2019 vs. shorter 2010-2019). All 6 test causes agree.

Significance threshold (α=0.05 vs. stricter α=0.01). All 6 test causes agree.

Baseline trend shape (linear vs. curved/quadratic). 2 cause(s) disagree: Diseases of heart, Cerebrovascular disease.

The trend-shape check is the one that matters: heart disease and cerebrovascular disease lose significance when the pre-pandemic baseline is allowed to curve instead of being forced into a straight line — part of what the linear method reads as a 2020 disruption for these two causes could instead be the natural curvature of their pre-existing trend. Diabetes, drug overdose, and cancer hold up across every axis tested and are the most robust of the 5 disrupted causes.

Separately, measured lag-1 autocorrelation of each cause's own pre-pandemic residuals is large for 5 of 6 causes (0.65-0.93; only cancer is low, at 0.19). The prediction-interval math assumes independent year-to-year residuals, which this data doesn't really satisfy — reported p-values throughout this report are likely more confident than a model accounting for this would produce. This does not overturn the results (deviations found are large, and the negative control still passed), but it is a real limitation, not a footnote.

Vintage-bridging reliability (D76 vs. D158 database overlap, 2018-2019): all causes within the 10% reliability threshold — median relative offset is 0% for every cause tested.

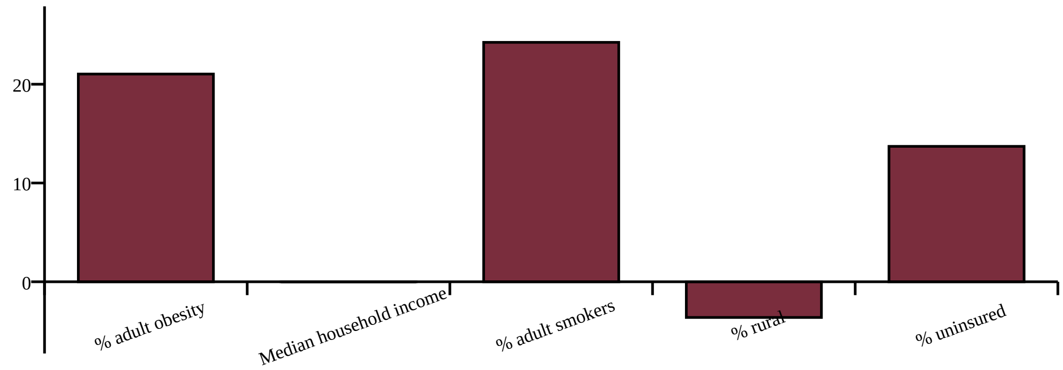
5. Which counties were hit hardest

For the two causes with real county-level data — diabetes and drug overdose, ~3,000 counties each, pre-period 2015-2019 vs. post-period 2020-2024 — disruption magnitude is regressed against real County Health Rankings & Roadmaps context variables. This stage uses crude rate, not age-adjusted rate, for both periods: CDC WONDER does not offer age-adjustment at county granularity for its 2018-2024 database, so part of any county's measured disruption could reflect its own population-aging trajectory rather than a COVID-era shift.

Diabetes mellitus — 5 of 5 context variables survive FDR correction

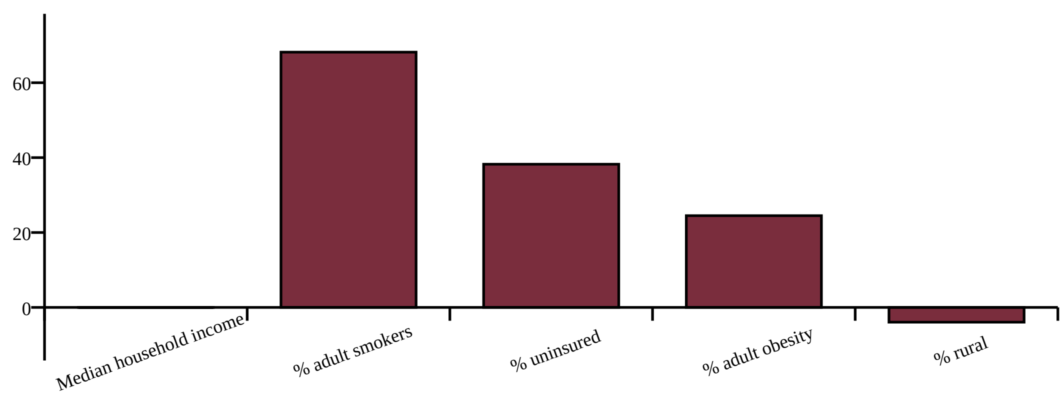
Context variable	Slope	p-value	FDR-sig.
% adult obesity	21.030	0.000545	Yes
Median household income	-0.000	0.000886	Yes

% adult smokers	24.239	0.00244	Yes
% rural	-3.614	0.00566	Yes
% uninsured	13.709	0.0128	Yes



Drug overdose — 4 of 5 context variables survive FDR correction

Context variable	Slope	p-value	FDR-sig.
Median household income	-0.000	3.68e-15	Yes
% adult smokers	68.159	8.36e-09	Yes
% uninsured	38.233	4.32e-06	Yes
% adult obesity	24.497	0.00409	Yes
% rural	-3.942	0.0688	No



Higher uninsured rate, smoking rate, and obesity rate all predict larger disruption for both causes. Higher median household income predicts smaller disruption for both. Higher rurality predicts smaller disruption for both — the one genuinely counterintuitive result, running against a common assumption that rural areas were hit hardest. These are associations, not causal claims.

Rurality finding: a real caveat, found on self-audit

Counties excluded from this regression (too few non-suppressed years) are far more rural, on average, than the counties included: Diabetes mellitus: excluded counties average 78% rural vs. 31% rural for counties included; Drug overdose: excluded counties average 74% rural vs. 24% rural for counties included. Splitting the included counties at their own median rurality shows the two causes are not equally trustworthy here.

Diabetes mellitus: the relationship holds up and even strengthens among the more-rural half ($p=0.00325$) vs. the less-rural half ($p=0.176$).

Drug overdose: the relationship is driven almost entirely by the less-rural half ($p=0.00016$) and is not significant among the more-rural half ($p=0.4$) — read with real skepticism.

6. Limitations

• Observational, ecological design — no individual-level causal inference. The county-level heterogeneity stage carries ecological fallacy risk by name: a county-average relationship does not necessarily hold at the individual level.

• Selection bias in the county-level heterogeneity sample: counties excluded by the suppression filter are disproportionately rural, and the rurality finding is more trustworthy for diabetes (holds up among more-rural included counties) than for drug overdose (driven by less-rural counties, not significant among more-rural ones) — see section 5.

• Mortality-vintage discontinuity between the two CDC WONDER databases (mitigated by bridging-overlap validation, not eliminated).

• ICD-10 coding practices may have shifted during 2020-2021 due to strain on death-certification systems, independent of true mortality changes.

• Cancer's pre-registered prior was an expected null result; the real result contradicts that — cancer shows a significant, still-persisting disruption, not the null originally expected.

• Temporal autocorrelation of baseline residuals is not modeled and is empirically large (0.65-0.93 for 5 of 6 test causes) — reported p-values are likely optimistic.

• “Significant” and “large” are different claims: cancer's disruption is real but small (+1.7% acute) next to heart disease, diabetes, cerebrovascular disease, or overdose (+26% to +41%).

• Heart disease and cerebrovascular disease's significance is not robust to the choice of linear vs. curved baseline trend shape.

• County-level heterogeneity uses crude rate, not age-adjusted rate (WONDER does not offer age-adjustment at county granularity for the 2018-2024 database), so measured disruption magnitude may partly reflect each county's own population-aging trajectory.

• Small-county suppression and instability at the county-level heterogeneity stage.

• PLACES model-based behavioral estimates (CHR&R smoking/obesity/inactivity).

• Context-variable vintage is post-period, not pre-period: all five heterogeneity-stage context variables come from CHR&R's 2024 release, measured during or after the 2020-2024 disruption window, not a pre-pandemic baseline — income and the uninsured rate plausibly moved during the pandemic itself.

• Multiple testing across both the 6-cause family and, separately, the context-variable family per cause.

• Spatial non-independence of counties (spatial autocorrelation not modeled).

• All findings remain associational — the mechanism behind any confirmed disruption (direct viral effect vs. deferred care vs. isolation vs. economic stress) cannot be separated by mortality data alone.

7. Data provenance and reproducibility

All national-level results use 15 real CDC WONDER exports (7 from the 1999-2019 database, 8 from the 2018-2024 database). All county-level heterogeneity results use 4 real CDC WONDER exports (diabetes and drug overdose, pre/post period, county-grouped). Nothing in this project is synthetic. Full methodology, pre-registered hypotheses, and every deviation logged as it happened: research_protocol.md. Exact manual data-export steps: manual_data_acquisition.md. Interactive app and full source: github.com/aidanc667/covid-mortality-disruption-lab.